



Even deers have COVID now!

New study suggests that the SARS-CoV-2 is spreading rapidly within deer populations.
<https://digit.in/nov21-35>



Vaccine for children

FDA authorizes Pfizer's COVID-19 Vaccine for children aged 5 to 11.
<https://digit.in/nov21-36>

Human perception of temperature and pressure



For a long time never knew why those snuggles and warmth of the fireplace felt so great!

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erception is something that we thrive on as humans. Given the amount of research conducted

every day in labs and test centers across the world, one might think that we know almost everything about our human body. As it turns out, we are a complex combination of cells and organs, and some unsolved mysteries remain. One such secret was that of how we perceive temperature and pressure. It was researchers like René Descartes, whose understanding and representation of our nervous system laid the foundation for the years of

research that was conducted in the field of neurosciences.

The two individuals behind the amazing discovery of the logic behind the human perception of temperature and pressure are the recipients of The Nobel Prize in Physiology or Medicine 2021, David Julius and Ardem Patapoutian. Before their work saw the light of day, we knew the working of the nervous system but did not know how the body converts the stimulus of temperature and pressure into electric nerve signals.

SENSING THE RISING MERCURY

David Julius and his team of researchers started their quest of working on the human perception of temperature in the 1990s at the University of California. Julius and his team used capsaicin to be the stimuli inducing compound. The effects of capsaicin during the consumption of chilli, characterised by the sensation of heat and burning, were already known before the start of the research.

In 1997, they published a research article titled, The capsaicin receptor: a

heat-activated ion channel in the pain pathway. Julius and his team elucidated their technique using which they understood how signals are activated in the human body after it comes in contact with capsaicin. They used a method where they created a massive library of DNA fragments, with genes that were expressed in the sensory neurons, which can react to pain, heat, and touch. From this library, as stated in the article, they were able to isolate “a functional cDNA encoding a capsaicin receptor from sensory neurons”. After further studies, as mentioned in the paper, the researchers were able to find that the receptor was activated by an increase in temperature, more so when it exceeded the painful range. The studies said this receptor was “structurally related to members of the TRP family of ion channels”.

The following year, the team made further developments, classifying a cloned vanilloid receptor, named VR1 which was also a cation channel, which was activated by heat. “...analysis of heat-evoked single-channel currents in excised membrane patches suggests that heat gates VR1 directly. We also show that protons decrease the temperature threshold for VR1 activation such that even moderately acidic conditions (pH $\approx$ 5.9) activate VR1 at room temperature. VR1 can therefore be viewed as a molecular integrator of chemical and physical stimuli that elicit pain,” wrote Julius and his team in their 1998 research article, The Cloned Capsaicin Receptor Integrates Multiple Pain-Producing Stimuli.

Numerous years of research followed, after which the team was able to single out an ion channel that was activated by painful heat, and it was named TRPV1. Following this discovery, several other avenues of research pertaining to similar receptors opened